vakint p-integrals: RustRed vs Forcer

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Reading guide. Each cell is the sum over all tops of that loop count at the given dots, times a numerator of that rank. Bold: RustRed cold (seconds, median). F: Forcer. ×: the ratio RustRed cold over Forcer, cold against cold because neither replays results; green is RustRed faster. ws, wc: RustRed warm on its own cache, and warm on one cache shared along the cells of that loop count. ✓: result exactly equal to Forcer's. n: integrals in the input. †: fewer repetitions than the protocol. Grey: Forcer only in v1. **Caveat:** indicative timings on a shared, loaded host (foreign load on node 4 not excluded): median foreign share of the measuring CPUs 1L n/a (before the sampler), 2L 0.82, 3L 0.12, 4L 0.52.

Performance gates. 4L on par with Forcer at 1 thread, better at 16: n/a in v1 (RustRed 4L: M3). Cold start dominated by loading: load 1.01 s (median). G10b, 3L rank ≥ 4, 16 threads, RustRed cold ≤ Forcer (recorded, not blocking): ×0.14 to ×1.03, 9 of 9 cells.



Table 1: 1 thread (`n_cores` 1, `form`); RustRed arithmetic `modular_polynomial`.

L	dots	rank 0	rank 2	rank 4	rank 6	rank 8
1 L	0	**0.006** ✓ F 0.442 **×0.01** ws 0.004 wc 0.006 n=1	**0.007** ✓ F 0.436 **×0.02** ws 0.004 wc 0.006 n=1	**0.007** ✓ F 0.440 **×0.01** ws 0.005 wc 0.007 n=1	**0.007** ✓ F 0.436 **×0.02** ws 0.004 wc 0.007 n=1	**0.007** ✓ F 0.440 **×0.02** ws 0.005 wc 0.007 n=1
	1	**0.006** ✓ F 0.438 **×0.01** ws 0.004 wc 0.007 n=1	**0.008** ✓ F 0.435 **×0.02** ws 0.005 wc 0.007 n=1	**0.009** ✓ F 0.438 **×0.02** ws 0.007 wc 0.008 n=1	**0.011** ✓ F 0.444 **×0.02** ws 0.008 wc 0.010 n=1	**0.014** ✓ F 0.450 **×0.03** ws 0.011 wc 0.013 n=1
	2	**0.006** ✓ F 0.434 **×0.01** ws 0.005 wc 0.006 n=1	**0.006** ✓ F 0.436 **×0.01** ws 0.005 wc 0.007 n=1	**0.007** ✓ F 0.437 **×0.02** ws 0.005 wc 0.006 n=1	**0.007** ✓ F 0.441 **×0.02** ws 0.005 wc 0.006 n=1	**0.007** ✓ F 0.441 **×0.02** ws 0.005 wc 0.006 n=1
2 L	0	**0.195** ✓ F 0.482 **×0.41** ws 0.006 wc 0.013 n=1	**8.43** ✓ F 0.493 **×17** ws 0.005 wc 0.196 n=1	**0.392** ✓ F 0.478 **×0.82** ws 0.006 wc 0.128 n=1	**0.203** ✓ F 0.489 **×0.42** ws 0.007 wc 0.286 n=1	**0.150** ✓ F 0.526 **×0.28** ws 0.008 wc 0.709 n=1
	1	**0.013** ✓ F 0.483 **×0.03** ws 0.006 wc 0.067 n=1	**0.126** ✓ F 0.501 **×0.25** ws 0.006 wc 0.363 n=1	**0.078** ✓ F 0.498 **×0.16** ws 0.006 wc 0.676 n=1	**0.075** ✓ F 0.488 **×0.15** ws 0.007 wc 0.549 n=1	**0.192** ✓ F 0.517 **×0.37** ws 0.008 wc 0.406 n=1
	2	**0.440** ✓ F 0.484 **×0.91** ws 0.007 wc 6.33 n=1	**0.393** ✓ F 0.494 **×0.80** ws 0.007 wc 0.520 n=1	**2.68** ✓ F 0.501 **×5.36** ws 0.007 wc 0.192 n=1	**0.023** ✓ F 0.504 **×0.04** ws 0.009 wc 0.536 n=1	**0.026** ✓ F 0.515 **×0.05** ws 0.011 wc 4.33 n=1
3 L	0	**0.566** ✓ F 1.66 **×0.34** ws 0.021 wc 0.076 n=3	**2.16** ✓ F 1.76 **×1.23** ws 0.028 wc 1.44 n=3	**0.286** ✓ F 2.01 **×0.14** ws 0.018 wc 0.288 n=3	**9.19** ✓ F 2.38 **×3.87** ws 0.079 wc 0.587 n=3	**1.04** ✓ F 2.82 **×0.37** ws 0.228 wc 0.986 n=3
	1	**3.48** ✓ F 1.93 **×1.80** ws 0.023 wc 0.090 n=3	**0.206** ✓ F 1.86 **×0.11** ws 0.030 wc 0.399 n=3	**1.37** ✓ F 2.02 **×0.67** ws 0.031 wc 0.784 n=3	**0.832** ✓ F 2.42 **×0.34** ws 0.134 wc 0.831 n=3	**1.60** ✓ F 2.88 **×0.56** ws 0.458 wc 1.40 n=3
	2	**0.345** ✓ F 1.96 **×0.18** ws 0.026 wc 0.135 n=3	**0.345** ✓ F 2.06 **×0.17** ws 0.036 wc 0.234 n=3	**1.47** ✓ F 2.21 **×0.66** ws 0.035 wc 0.344 n=3	**1.45** ✓ F 2.59 **×0.56** ws 0.123 wc 0.888 n=3	**1.28** ✓ F 3.32 **×0.38** ws 0.385 wc 1.12 n=3
4 L	0	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11
	1	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11
	2	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11	n/a (M3) F — n=11
phys.		2 L **0.357** ✓ F 0.519 **×0.69** ws 0.022 wc 0.023 n=11	3 L **0.712** ✓ F 1.34 **×0.53** ws 0.402 wc 0.567 n=53	4 L n/a (M3) F — n=293	5L canary: n/a	

Table 2: 16 threads (`n_cores` 16, `tform -w16`); RustRed arithmetic `modular_polynomial`.

L	dots	rank 0	rank 2	rank 4	rank 6	rank 8
1 L	0	**0.010** ✓ F 0.467 **×0.02** ws 0.007 wc 0.012 n=1	**0.012** ✓ F 0.458 **×0.03** ws 0.007 wc 0.011 n=1	**0.013** ✓ F 0.462 **×0.03** ws 0.007 wc 0.011 n=1	**0.014** ✓ F 0.475 **×0.03** ws 0.008 wc 0.022 n=1	**0.013** ✓ F 0.494 **×0.03** ws 0.007 wc 0.015 n=1
	1	**0.015** ✓ F 0.467 **×0.03** ws 0.006 wc 0.043 n=1	**3.97** ✓ F 0.454 **×8.76** ws 0.015 wc 0.019 n=1	**0.033** ✓ F 0.479 **×0.07** ws 0.020 wc 0.023 n=1	**0.038** ✓ F 0.486 **×0.08** ws 0.026 wc 0.029 n=1	**0.045** ✓ F 0.500 **×0.09** ws 0.033 wc 0.036 n=1
	2	**0.009** ✓ F 0.473 **×0.02** ws 0.007 wc 0.012 n=1	**0.666** ✓ F 0.460 **×1.45** ws 0.008 wc 0.016 n=1	**0.017** ✓ F 0.468 **×0.04** ws 0.007 wc 1.08 n=1	**0.012** ✓ F 0.459 **×0.03** ws 0.008 wc 0.010 n=1	**0.016** ✓ F 0.479 **×0.03** ws 0.009 wc 0.010 n=1
2 L	0	**0.025** ✓ F 0.671 **×0.04** ws 0.011 wc 0.027 n=1	**0.845** ✓ F 0.594 **×1.42** ws 0.016 wc 0.521 n=1	**0.745** ✓ F 0.590 **×1.26** ws 0.010 wc 0.024 n=1	**0.579** ✓ F 0.630 **×0.92** ws 0.027 wc 0.757 n=1	**0.636** ✓ F 0.599 **×1.06** ws 0.012 wc 0.400 n=1
	1	**0.128** ✓ F 0.547 **×0.23** ws 0.013 wc 0.242 n=1	**0.021** ✓ F 0.575 **×0.04** ws 0.012 wc 10.0 n=1	**1.03** ✓ F 0.567 **×1.82** ws 0.011 wc 0.146 n=1	**0.149** ✓ F 0.576 **×0.26** ws 0.010 wc 9.06 n=1	**0.531** ✓ F 0.579 **×0.92** ws 0.012 wc 0.570 n=1
	2	**0.102** ✓ F 0.487 **×0.21** ws 0.011 wc 0.021 n=1	**0.025** ✓ F 0.611 **×0.04** ws 0.010 wc 0.872 n=1	**0.023** ✓ F 0.608 **×0.04** ws 0.013 wc 0.219 n=1	**0.035** ✓ F 0.547 **×0.06** ws 0.013 wc 5.07 n=1	**0.615** ✓ F 0.577 **×1.07** ws 0.017 wc 0.634 n=1
3 L	0	**0.127** ✓ F 1.80 **×0.07** ws 0.024 wc 0.215 n=3	**0.630** ✓ F 1.95 **×0.32** ws 0.034 wc 0.505 n=3	**0.299** ✓ F 2.16 **×0.14** ws 0.022 wc 2.83 n=3	**0.523** ✓ F 2.50 **×0.21** ws 0.097 wc 0.349 n=3	**0.653** ✓ F 2.77 **×0.24** ws 0.249 wc 0.847 n=3
	1	**0.061** ✓ F 2.08 **×0.03** ws 0.028 wc 0.065 n=3	**0.455** ✓ F 1.99 **×0.23** ws 0.038 wc 0.665 n=3	**1.03** ✓ F 2.18 **×0.47** ws 0.036 wc 0.134 n=3	**0.476** ✓ F 2.55 **×0.19** ws 0.161 wc 0.496 n=3	**1.08** ✓ F 2.88 **×0.37** ws 0.466 wc 1.03 n=3
	2	**0.070** ✓ F 2.11 **×0.03** ws 0.032 wc 6.58 n=3	**0.409** ✓ F 2.22 **×0.18** ws 0.045 wc 0.253 n=3	**2.45** ✓ F 2.37 **×1.03** ws 0.046 wc 5.66 n=3	**0.424** ✓ F 2.67 **×0.16** ws 0.164 wc 0.621 n=3	**0.908** ✓ F 3.06 **×0.30** ws 0.438 wc 5.06 n=3
4 L	0	n/a (M3) F 21.4 n=11	n/a (M3) F 36.9 n=11	n/a (M3) F 47.5 n=11	n/a (M3) F 70.1 n=11	n/a (M3) F 95.3 n=11
	1	n/a (M3) F 38.6 n=11	n/a (M3) F 45.0 n=11	n/a (M3) F 52.4 n=11	n/a (M3) F 76.6 n=11	n/a (M3) F 98.0 n=11
	2	n/a (M3) F 45.4 n=11	n/a (M3) F 51.9 n=11	n/a (M3) F 65.2 n=11	n/a (M3) F 100 n=11	n/a (M3) F 132 n=11
phys.		2 L **0.175** ✓ F 0.628 **×0.28** ws 0.062 wc 0.070 n=11	3 L **0.779** ✓ F 1.24 **×0.63** ws 0.612 wc 0.766 n=53	4 L n/a (M3) F 65.4 n=293	5L canary: n/a	

Table 3: RustRed cold, arithmetic `exact` against `modular_polynomial` (mod.); gain = exact / mod.

cell	1 thread			16 threads		
	exact	mod.	gain	exact	mod.	gain
L1 d0 r0	0.006	0.006	**×1.04**	0.230	0.010	**×24**
L1 d1 r4	2.41	0.009	**×261**	0.014	0.033	**×0.44**
L1 d2 r8	0.007	0.007	**×0.92**	0.744	0.016	**×47**
L2 d0 r0	0.148	0.195	**×0.76**	0.012	0.025	**×0.49**
L2 d1 r4	0.014	0.078	**×0.18**	1.39	1.03	**×1.34**
L2 d2 r8	0.023	0.026	**×0.86**	0.618	0.615	**×1.01**
physics L2	4.31	0.357	**×12**	0.043	0.175	**×0.24**
L3 d0 r0	0.078	0.566	**×0.14**	0.645	0.127	**×5.08**
L3 d1 r4	0.951	1.37	**×0.70**	5.93	1.03	**×5.79**
L3 d2 r8	4.68	1.28	**×3.67**	1.02	0.908	**×1.12**
physics L3	0.748	0.712	**×1.05**	9.39	0.779	**×12**

Table 4: Reuse, `modular_polynomial`: warm time over cold time per loop count (sums over the cells). ws clean: warm-same children with 0 misses and 0 inserts. wc: the chain's cache counters at 16 threads.

L	1 thread		16 threads		ws clean	wc hits / misses
	ws/cold	wc/cold	ws/cold	wc/cold		
1 L	×0.70	×0.93	×0.04	×0.28	90 / 90	336 / 67
2 L	×1/107	×1.11	×0.05	×5.06	96 / 96	1207 / 1042
3 L	×0.08	×0.39	×0.24	×2.52	96 / 96	78418 / 51148

Sub-lines (seconds, sum over the cells measured at 1 thread).

	1 L	2 L	3 L	4 L
RustRed reduction only	0.016	0.118	7.14	—
RustRed load (excluded)	15.1	17.0	15.6	—
Forcer FORM subprocess	6.18	7.50	34.4	—
Raw FORM reference	6.36	7.94	34.7	—

Method. Timed: the wall time of `Vakint::evaluate` in Retain mode (exact reduction onto masters, same frontend and finalizer for both backends) in one fresh process per measurement, with the input parsed, the engine and library loaded and the 1L bubble with four dots (in no cell: vakint memoises Forcer's FORM runs per process) evaluated once beforehand, by either backend, on directories of its own; repetitions: 3, median. RustRed cold: empty ordered-cache directory, shipped caches off. ws: a fresh process on a read-only copy of the cache of its cold run. wc: per loop count, one chain over the cells in table order sharing one cache from empty, each cell measured once in the chain. Forcer cold: private empty materialisation directory; Forcer warm: materialised. Forcer has no result cache. The raw reference runs the generated FORM programs directly, one process per top. OS pages are not flushed. 4L and the 4L physics cell: Forcer only, 16 threads, warm, one repetition, capped per cell. Physics row: massless-quark (L-1)-rung gluon self-energy, Feynman gauge, transverse projector, external momentum kept.

Provenance. Run `2026-11-08T18:42:02Z-84a7c0dee556`, profile release, `FORM 5.0.0 (Jan 1 1980)`, Forcer 68a30bf, AMD EPYC 9754 128-Core Processor on itphlies, CPUs 1 thread: 128; 16 thread: 128-143, binding `--cpunodebind=4 --membind=4`, RustRed arithmetic `modular_polynomial` in Tables 1, 2 and 4, `exact` in Table 3 (each child records the arithmetic it resolved), `RUSTRED_ORDERED_ADJOINT_TAIL` unset, no official external data loaded, `1774` measurements in `data/p_results.json`.